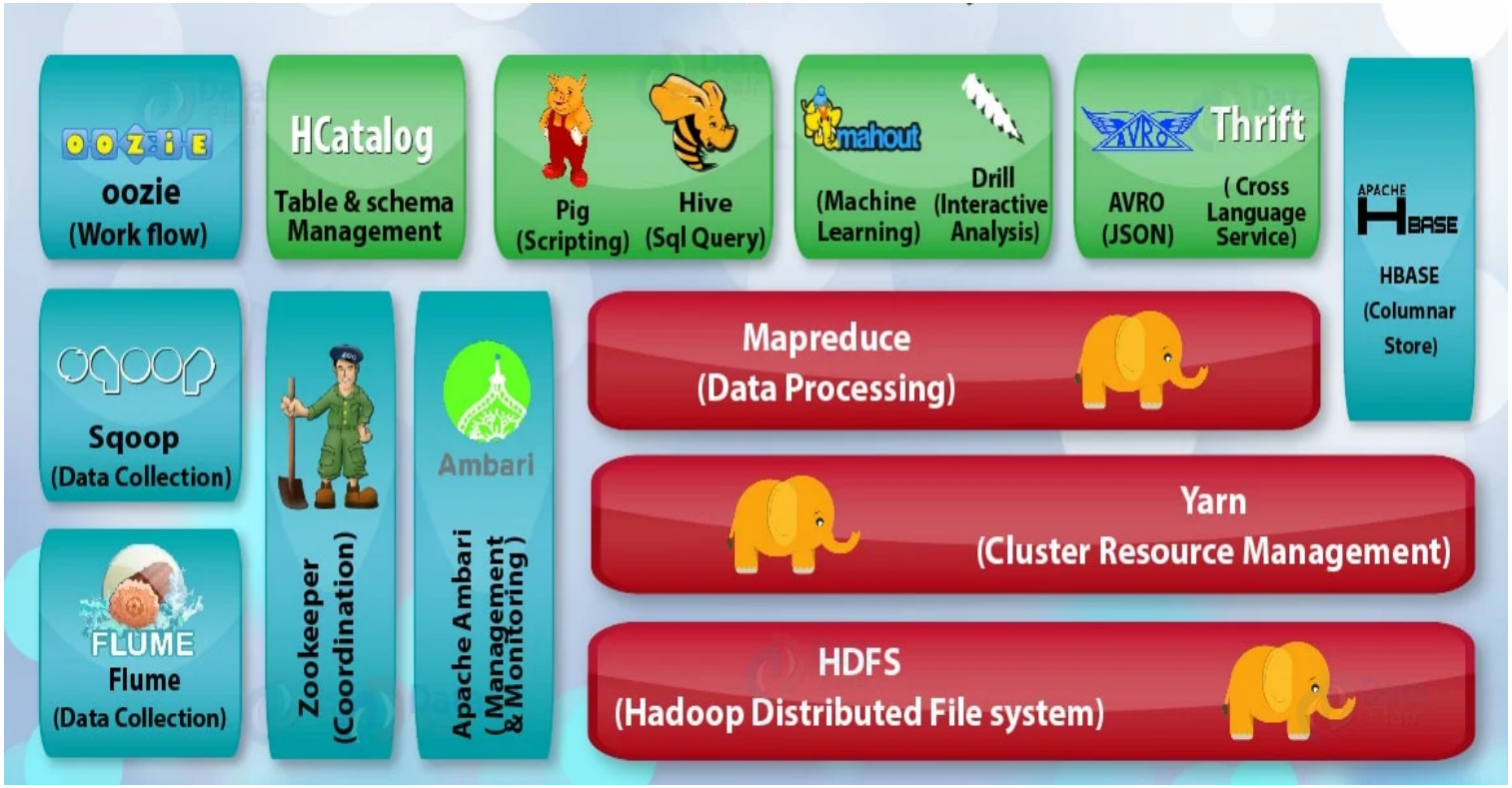
IT Workshop II (Cloud Computing): Hadoop - Concepts

September 27, 2026

Hadoop - Overview

- Apache Hadoop is an open source framework for distributed batch processing of big data.
- Hadoop ecosystem consists of a number of projects. Some of them are given below-
  - **HDFS** - It is a distributed file system that runs on large clusters and provides high throughput access to data.
  - **Hadoop MapReduce** - It is a parallel programming model that allows distributing large scale computations in a sequence of operations on data sets of key-value pairs.
  - **Hadoop YARN** - A framework for job sequencing and clustering resource management.
  - **HBase** - It is scalable, non-relational, distributed, column-oriented database.
  - **Zookeeper** - It is a high performance distributed coordination service for maintaining configuration information, naming etc.

Hadoop Ecosystem

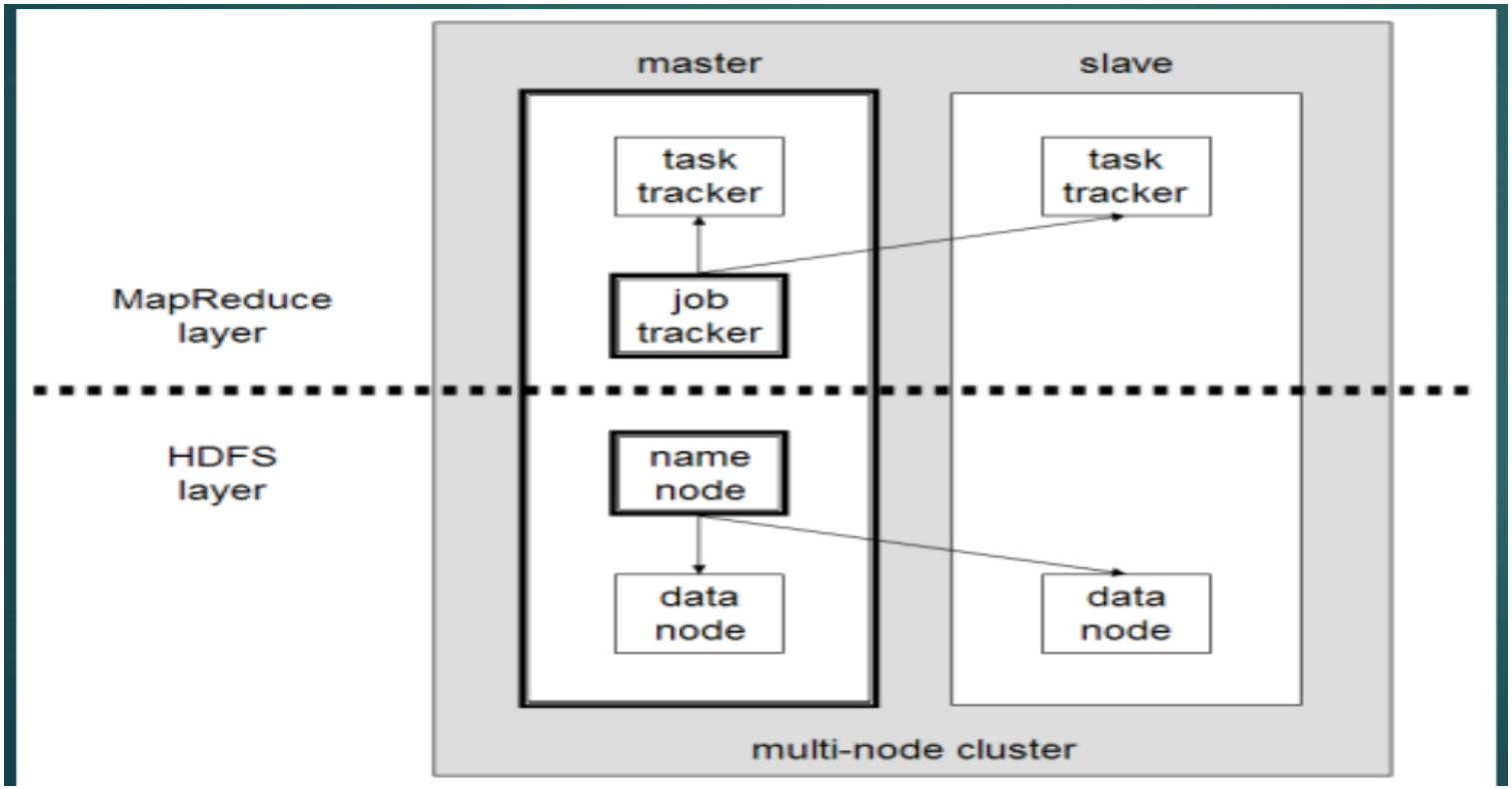


Source: Internet

HDFS

- Hadoop Distributed File System (HDFS) is designed to reliably store very large files across machines in a large cluster. It is inspired by the GoogleFileSystem.
- Distribute large data file into blocks.
- Blocks are managed by different nodes in the cluster.
- Each block is replicated on multiple nodes.
- Name node stores metadata information about files and blocks.

Multi-Node Hadoop Cluster



Nodes

- **NameNode:**
  - Master of the system
  - Maintains and manages the blocks which are present on the DataNodes
- **DataNodes:**
  - Slaves which are deployed on each machine and provide the actual storage.
  - Responsible for serving read and write requests for the clients.
- **JobTracker:** Takes care of all the job scheduling and assign tasks to Task Trackers.
- **TaskTracker:** A node in the cluster that accepts tasks - Map, Reduce and Shuffle operations - from a JobTracker.

Other Hadoop Components in Ecosystem

- **HBase:** Hadoop database for random read/write access.
- **Hive:** SQL-like queries and tables on large datasets.
- **Pig:** Data flow language and compiler.
- **Sqoop:** Integration of databases and data warehouses with Hadoop.
- **ZooKeeper:** Coordination service for distributed applications.

Hbase

- HBase is an open source, non-relational, distributed database modeled after Google's BigTable.
- It runs on top of Hadoop and HDFS, providing BigTable-like capabilities for Hadoop.
- It is a type of NoSql database.
- HBase supports an easy to use Java API for programmatic access.
- HBase also supports REST for non-Java front-ends.

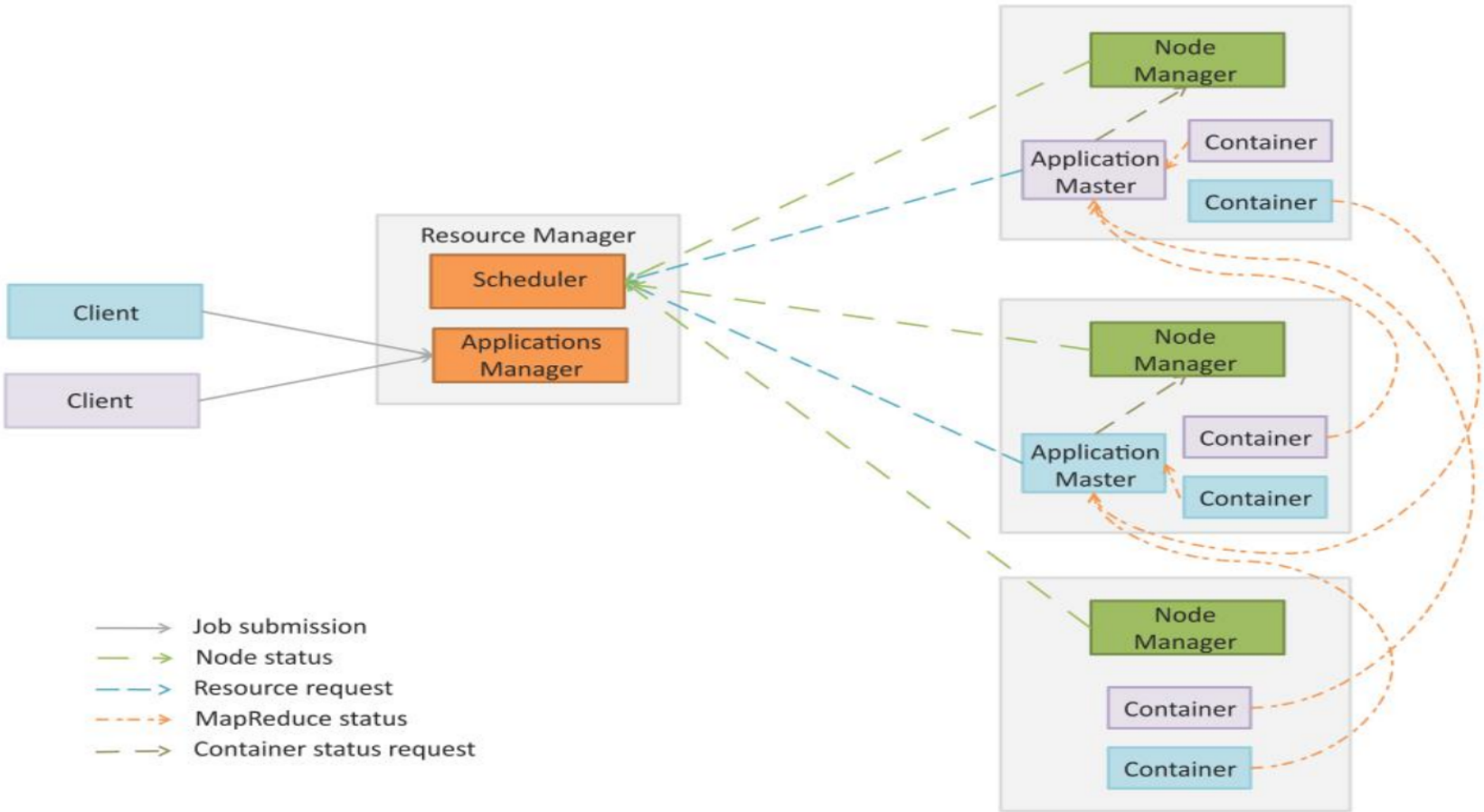
Difference between HDFS and Hbase

HDFS	Hbase
Good for storing large file	Built on top of HDFS. Good for hosting very large tables like billions of rows X millions of column
Write once. Append to files in some of recent versions but not commonly used	Read/write many
No random read/write	Random read/write
No individual record lookup rather read all data	Fast records lookup(update)

YARN Components

- YARN is an operating system for Hadoop that supports different processing engines on a Hadoop cluster such as MapReduce for batch processing, Apache Storm for stream processing, etc.
- **Resource Manager (RM):** RM manages the global assignment of compute resources to applications.
- **Application Master (AM):** A per-application AM manages the application's life cycle.
- **Node Manager (NM):** A per-machine NM manages the user processes on that machine.
- **Containers:** Container is a bundle of resources allocated by RM (memory, CPU, network, etc.)

YARN Components



Hadoop Schedulers

- Hadoop scheduler is a pluggable component that makes it open to support different scheduling algorithms.
- The default scheduler in Hadoop is FIFO.
- Two advanced schedulers are also available - the Fair Scheduler, developed at Facebook, and the Capacity Scheduler, developed at Yahoo.
- The pluggable scheduler framework provides the flexibility to support a variety of workloads with varying priority and performance constraints.

The End